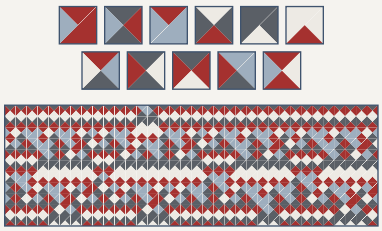
COMPUTING WITH TILES

Wang tiles that add, and a wall that counts in Fibonacci numbers

I. WANG TILES

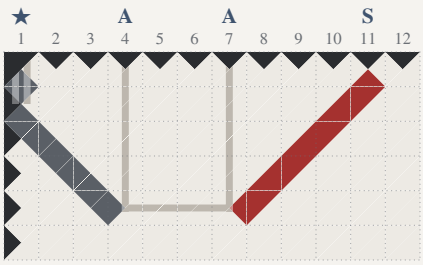
A *Wang tiling* is a tiling of square tiles with labels on each edge, such that the labels on touching edges match. Tiles may not be rotated or reflected. We represent labels as a combination of color and etched lines, which must form a continuous line on a uniform background where edges meet. Hao Wang (王浩)[†] conjectured in 1961 that any set of such tiles that can cover the plane can also cover it periodically. Robert Berger (1966) showed otherwise: tiles can mimic any computer program (they are *Turing complete*), so whether a set tiles the plane is undecidable. Along the way he found the first *aperiodic* set, 20,426 tiles that tile the plane but never repeat. At right is the smallest aperiodic set possible: 11 tiles with 4 unique edge labels (Jeandel & Rao, 2015).



a minimal aperiodic set, and a patch of its tiling

Our goal is to achieve an aperiodic tiling for the home using tiles in easily obtainable practical shapes. Wang tiles are attractive because the aperiodicity is from edge matching rather than shape. The Jeandel & Rao tile set thus achieves our goal, but let's go further.

II. TILES THAT ADD



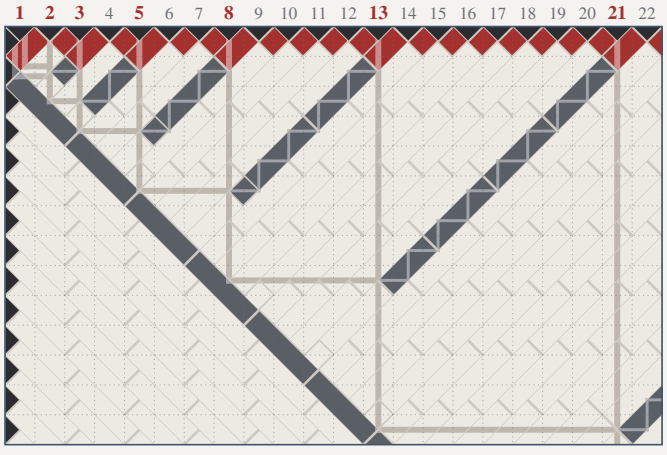
A at 4 and 7 force S to 11



This set of 16 tiles with 6 labels demonstrates arithmetic with Wang tiles. We will work on a quarter plane whose top and left edges are constrained to be black. Place the **A** tiles on the top row at your input values, A_1 and A_2. The ★ tile can only be placed in the corner. The tile edge constraints cause an etched line to descend from each **A**. A charcoal diagonal leaves ★; where it meets the first line it turns and starts a horizontal line at a depth corresponding to A_1. When that meets the second line a red diagonal climbs up. This forces the **S** tile, representing the sum, to be placed at column $A_1 + A_2$.

III. FIBONACCI NUMBERS





as it could be laid in 2″ × 8″ subway tile

By extending the previous tiles we can generate the Fibonacci numbers, each the sum of the two before. This set of 18 Wang tiles have 9 distinct edge labels; top and left borders are constrained to be black as before. The **F′** and **F″** tiles in the corner mark the first two numbers, 1 and 2.[‡] Every **F** sends an etched line down. A long charcoal diagonal leaves the corner, and where it meets the line of the **F** at column n, at depth n, the line turns and runs right to the descending line of the next **F**, at column m. There a charcoal staircase climbs back up, rising n rows as it moves n columns right, and reaches the top at column $m + n$: the next **F**, which then sends its own ray down.

The tiling is aperiodic because the Fibonacci numbers spread out, so this is another tiling which would achieve our goal. Using subway tiles oriented differently above and below the long charcoal diagonal adds visual interest.

[†] In Cantonese, Wang's surname 王 is a homophone of 黃, Wong: a name held by several members of this home.

[‡] **F′** and **F″** fix the start: with two more tiles that let **F″** stand at column 3 instead, the same machinery produces the Lucas numbers 1, 3, 4, 7, 11, 18, 29.

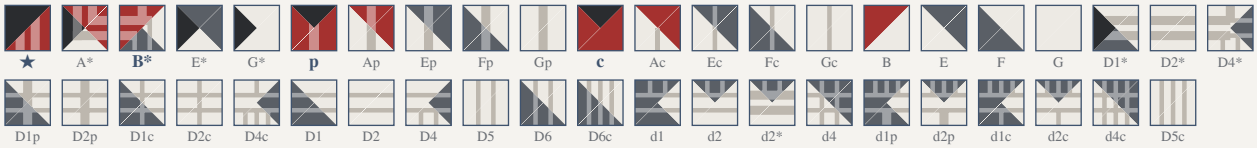
Addition and Fibonacci tiles after Grünbaum & Shephard, Tilings and Patterns, §11.4.

C. Scott Ananian · October 2026

A WALL THAT FINDS PRIMES

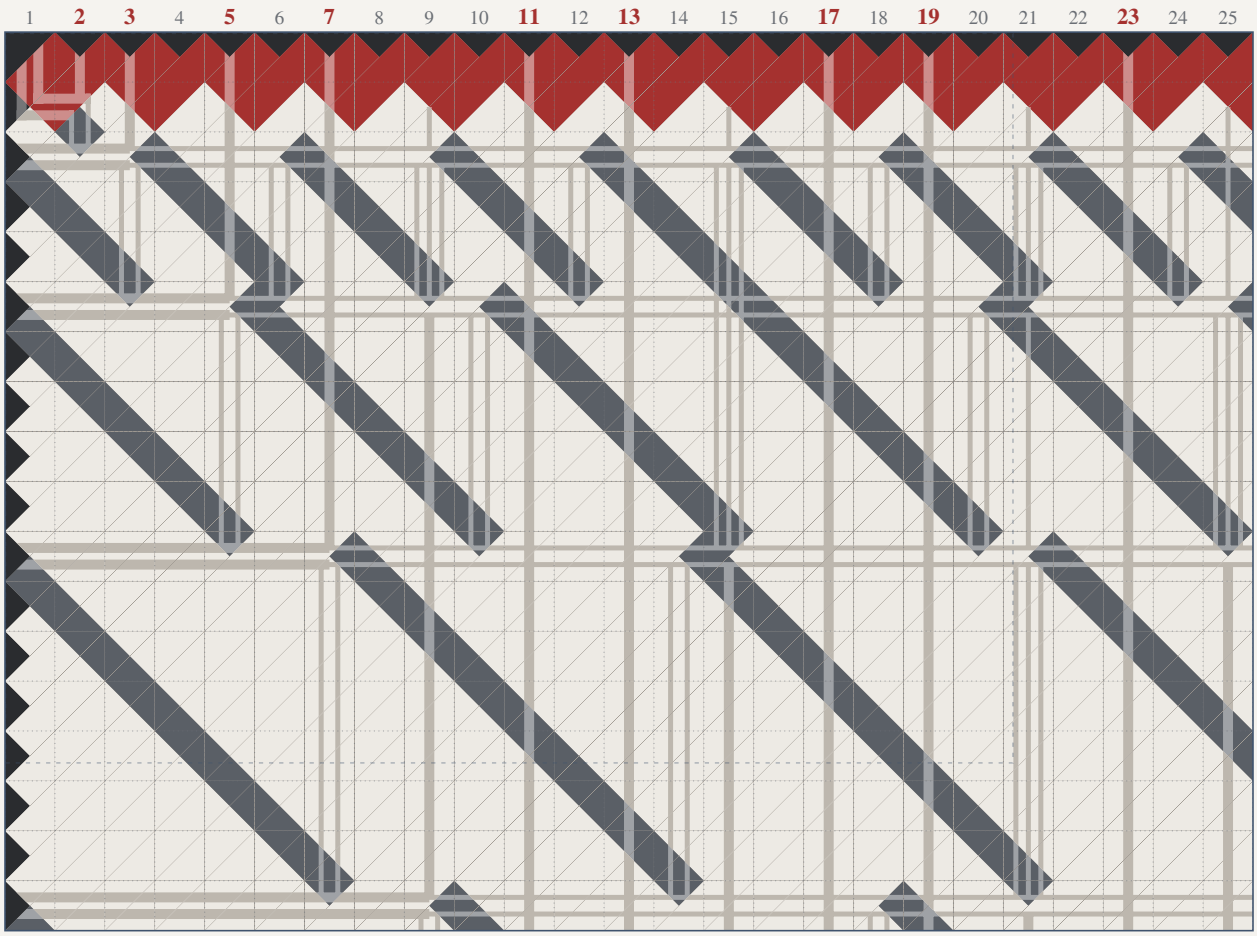
Wang tiles that sieve for primes, in ordinary 2″ × 8″ subway tile

IV. A PRIME NUMBER SIEVE



The wall in this room is an aperiodic tiling using a set of 43 Wang tiles with 15 distinct edge labels. Each edge label is shown by a tile color plus a pattern of etched lines, and a run of matching colors is laid as a single subway tile set on the diagonal. Tiles can not be rotated or reflected, edges must match, and the top and left borders must be black. Given these constraints, the tiles fill the infinite quarter plane in exactly one way, and that tiling is aperiodic because the gaps between primes never repeat: if they did, with a cycle adding up to P, then $p + P$, $p + 2P$, … would all be prime, but $p + pP = p(1 + P)$ is not.

Column n stands for the number n. Each odd column sends down an etched line which could be thin or thick. These interact with the thick double etched lines to start an vertical sequence of boxes of increasing odd sizes (3, 5, 7, 9, …) which then interact with the charcoal diagonals to repeat in rows. The line from the odd column is thick when it emerges from the first box in its row, but can remain thick only if it never crosses the vertical edge of another box; otherwise it becomes thin. (Even numbers get no ray; 2 has a tile of its own, **B***.) The thick rays reach the red top row at 2, 3, 5, 7, 11, 13, 17 and 19, marked by tile **p**. Composite numbers use tile **c**.



the Wang tiling, columns 1–25 and the top 18 rows; the wall in this room is the first 20 columns and 15 rows (dashed)

Tile set PRIME2b from github.com/cscott/tile-hack (wangprime.js).

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